INSTITUTE OF LAW, POLITICS AND DEVELOPMENT

Scuola Superiore Sant'Anna

Data Analysis in R using Agentic AI Assistants
18h - 3CFU

Prof. Kevin Koehler

Sant'Anna School of Advanced Studies
kevin.koehler@santannapisa.it

Course description

This course is an introduction to data analysis for the social sciences using R. It focuses on hands-on applications, rather than statistical or mathematical background. Basic knowledge of statistics is useful, but not strictly necessary, and no prior programming experience is assumed. In the first part of the course, students will gain familiarity with RStudio, a free user interface for R. In particular, we will cover basic issues of data management and data wrangling using the `tidyverse` family of tools, as well as data visualization using `ggplot` and the basics of R programming. In the second part of the course, we will then move on to data analysis. We will learn how to implement linear and logistic regressions in R, as well as how to efficiently present results in graphical or tabular form. In order to apply these tools, we will read a published paper, critically discuss the research design, and then replicate the analysis.

Throughout, we will work with **Agentic AI Assistants** as a standard part of the analytical workflow. These tools have made writing code the easy part. An assistant will produce a working `tidyverse` pipeline or a fitted regression model faster than you could look up the syntax, and it will do so whether or not the operation makes any sense for your data or your question. The assistant we use in this course is **OpenCode** (opencode.ai/download), an open-source agentic coding assistant available as a desktop app, in the terminal, or as an editor extension. OpenCode is free to install and comes with free models, so no paid AI subscription is required; participants who already have access to a commercial model (Claude, GPT, Gemini, or others) can connect it instead. Because the field is changing rapidly, the models available through OpenCode may change before the course starts; any adjustments will be communicated in advance.

This changes what is difficult about quantitative work, and the course is built around that change. Once producing an analysis costs almost nothing, the binding constraint is no longer whether you can write the code but whether you understand what the analysis does and can judge whether it is technically correct. Neither of those can be delegated to the tool that produced the output. An assistant will not tell you that a join silently duplicated half your observations, that a missing-value code entered the model as a real number, or that the coefficient it has just described in causal language rests on data that cannot support the claim. The code runs, the table renders, and nothing signals a problem. Recognizing such failures requires knowing what the operation was supposed to do in the first place. That is what we spend the course building: enough command of R and of the underlying statistics to direct these tools productively, to read what they give back, and to be able to say—of any result you put your name to—why it is right.

The textbook used in this class is Kennedy, Ryan and Philip D. Waggoner. 2021. *Introduction to R for Social Scientists: A Tidy Programming Approach*, CRC Press. Note that the textbook predates the tools discussed above; we will use it for the R substance and handle the AI workflow in class.

Learning objectives

By the end of the course, participants will be able to:

- Manage and wrangle social science data in R using the `tidyverse`, and verify that each transformation did what was intended.
- Produce and refine data visualizations with `ggplot`, and recognize when a plot is technically correct but misleading.
- Read and explain R code they did not write, including code generated by an AI assistant.
- Direct Agentic AI Assistants to carry out data management, visualization and analysis tasks, specifying those tasks precisely enough to get usable output.
- Detect the silent failures characteristic of AI-generated analysis: rows duplicated by a join, missing-value codes read as real observations, model assumptions left unstated.
- Estimate and interpret linear and logistic regression models, and distinguish a defensible reading of the coefficients from an overstated one.
- Present results in tabular and graphical form suitable for a research paper.
- Replicate a published quantitative analysis and account for divergences from the original results.
- Take responsibility for AI-assisted work: explain any line of submitted code and justify the analytical choices behind it.

Working with AI coding assistants

A note on how these tools are used here, since the approach differs from simply allowing them.

Agentic AI Assistants can read your files, propose and execute code, and iterate on errors without being asked each time. Used well, they remove most of the friction that normally makes introductory R painful—syntax errors, forgotten argument names, hunting through documentation. Used badly, they produce analyses that run cleanly and are wrong.

Each session below therefore pairs its R content with an AI component that follows the same pattern: use the tool to do the work, then verify the result against something independent—the raw data, the published table, your own reading of the output. Students are expected to use these tools in all practical work, and to be able to explain any line of code they submit.

Prerequisites

- Install R (here) and RStudio (here) before the first class
- Install OpenCode (here) before the first class. The desktop app is the simplest option for most participants. No account or paid subscription is needed; we will choose a model and connect OpenCode to your R project together in the first session.
- No prior programming experience is assumed.

Assessment

The course is graded **pass/fail**. The final deliverable is a **code file** (an R script or Quarto document) that implements a short analysis of the participant's choice, using the tools covered in class. It is due on the final day of class, **Monday 14 December 2026**.

Participants may use Agentic AI Assistants to write the code, but must be able to explain it: what each step does, why it is there, and how they checked that the result is correct. To pass, participants discuss their code with the instructor and answer questions about selected lines and analytical choices.

Overview of sessions

#	Date	Time	Topic
1	Fri 23 Oct	16:00–18:00	Introduction: Getting set up
2	Fri 30 Oct	16:00–18:00	Data management
3	Mon 2 Nov	09:00–11:00	Visualization
4	Mon 9 Nov	16:00–18:00	R programming
5	Mon 16 Nov	16:00–18:00	Exploring data
6	Mon 23 Nov	16:00–18:00	Linear regression
7	Mon 30 Nov	09:00–11:00	Replication I: U.S. Soft-Power and Foreign Policy Behavior (OLS)
8	Mon 7 Dec	16:00–18:00	Regression models for binary outcomes
9	Mon 14 Dec	16:00–18:00	Replication II: Does Counterbalancing Prevent Military Coups? (Logistic Regression)

Detailed description

Session 1: Getting set up Friday 23 October 2026, 16:00–18:00

- **Aim**: What is R and why should you care about it? What can you expect from this course (and what can't you)?
- **AI component**: Setting up OpenCode, choosing a model, and running it alongside RStudio. What an agentic assistant actually does when it reads your project, writes a script, and runs it. First exercise: have the tool create a project structure and explain each file back to you—then check whether the explanation is accurate. We also establish the working rule for the course: you may generate any code you like, but you must be able to say what it does.
- **Readings**: Kennedy and Waggoner 2021, Introduction and Chapter 2.

Session 2: Data management Friday 30 October 2026, 16:00–18:00

- **Aim**: Data management is a major part of quantitative work in the social sciences. In this session, we work with the `tidyverse` set of tools to master various aspects of data management and data wrangling.
- **AI component**: Prompting for a wrangling pipeline, then reading it back verb by verb. Wrangling is where AI assistants fail most quietly, so we practise the standard defences: check row counts before and after every join, inspect what a filter actually dropped, confirm that a recode covered every category. We look at a many-to-many join that silently inflates a dataset and at how to catch it.
- **Readings**: Kennedy and Waggoner 2021, Chapter 3.

Session 3: Visualization Monday 2 November 2026, 09:00–11:00

- **Aim**: R includes a range of powerful tools for data visualization. This session introduces the grammar of graphics approach implemented in the `ggplot` package.
- **AI component**: Iterating on a plot through successive prompts, which is where these tools are genuinely strong. The verification task is different here: the failure mode is not a broken plot but a misleading one. We examine what `ggplot` drops without warning, what overplotting conceals, and how a plot can be technically correct and still misrepresent the data.
- **Readings**: Kennedy and Waggoner 2021, Chapter 4.

Session 4: R programming Monday 9 November 2026, 16:00–18:00

- **Aim**: R is not only a software package for data analysis, but primarily a programming language. This session introduces the basics of R programming.

- **AI component**: This is the session where reading code you did not write becomes the explicit skill. Students specify a function, have the assistant write it, then design test cases—including edge cases the tool did not consider—and debug what fails. We also look at what happens when you ask the assistant to fix its own error, and when that goes in circles.

- **Readings**: Kennedy and Waggoner 2021, Chapter 5.

Session 5: Exploring data Monday 16 November 2026, 16:00–18:00

- **Aim**: Exploring your data is an essential first step preceding data analysis. This session introduces tools and procedures for data exploration in R.

- **AI component**: Using the assistant to produce a descriptive overview of an unfamiliar dataset quickly, then auditing it against the codebook. Particular attention to missing data: sentinel values such as −99 or 999 are routinely treated as real observations, which quietly corrupts every statistic downstream.

- **Readings**: Kennedy and Waggoner 2021, Chapter 6.

Session 6: Linear regression Monday 23 November 2026, 16:00–18:00

- **Aim**: Understanding the logic of linear regression is essential for understanding more complex regression approaches. This session discusses the linear regression model and its implementation in R.

- **AI component**: Fitting a model and producing a publication-quality regression table takes minutes with an assistant. Interpretation is the harder part. Students write their own reading of the coefficients before asking the tool for its interpretation, then compare. The recurring problem is causal language attached to observational estimates—these tools will tell you an effect exists on data that cannot support the claim.

- **Readings**: Kennedy and Waggoner 2021, Chapter 7 (specifically focus on section 7.5 Ordinary Least Squares Regression).

Session 7: Replication I - U.S. Soft-Power and Foreign Policy Behavior (OLS) Monday 30 November 2026, 09:00–11:00

- **Aim**: In this session we discuss the research design and replicate the analysis in an article by Benjamin Goldsmith and Yusaku Horiuchi (2012) published in *World Politics*. The article examines whether U.S. 'soft-power' matters for the foreign policy behavior of other states.

- **AI component**: A full AI-assisted replication, and a good test of the workflow, because we know what the answer should be. The assistant helps map codebook variables onto the paper's specifications; the published table is the ground truth. Where our numbers diverge from theirs, we work out why—and distinguish our errors from ambiguities in the original.

- **Readings**: Goldsmith, Benjamin E. and Yusaku Horiuchi. 2012. "In Search of Soft Power: Does Foreign Public Opinion Matter for US Foreign Policy?" *World Politics*, 64 (3): 555-585. doi:10.1017/S0043887112000123

- Replication data can be found here

Session 8: Regression models for binary outcomes Monday 7 December 2026, 16:00–18:00

- **Aim**: Many outcomes we are interested in as social scientists are binary: Whether an event does or does not occur (for example, civil war onset, military coups, revolutions, etc). This session discusses how the linear regression approach discussed before can be generalized to deal with such outcomes.

- **AI component**: Logistic regression is the clearest case in the course of an assistant producing correct code and incorrect interpretation. Log-odds, odds ratios and predicted probabilities are routinely conflated, and the resulting sentence sounds authoritative. We switch between several models in OpenCode and compare their readings of the same regression against our own, and settle on predicted probabilities as the quantity worth reporting.

- **Readings**: Kennedy and Waggoner 2021, Chapter 7 (specifically focus on section 7.6 Binary Response Models).

Session 9: Replication II - Does Counterbalancing Prevent Military Coups? (Logistic Regression) Monday 14 December 2026, 16:00–18:00

- **Aim**: In this session we discuss the research design and replicate the analysis in an article by Erica de Bruin (2017) published in the *Journal of Conflict Resolution*. The article examines whether counterbalancing is an effective strategy to prevent military coups.
- **AI component**: A second full replication, run with more independence than the first. We close with the practical question the course has been building toward: given tools that write the code for you, what does competent quantitative work now consist of, and what should you be willing to put your name to?
- **Readings**: De Bruin, Erica. 2018. "Preventing Coups d'état: How Counterbalancing Works." *Journal of Conflict Resolution* 62 (7): 1433–58. doi: 10.1177/0022002717692652.
- Replication data can be found at the journal website.